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EXAMINATION PAPER

Academic Year :	2025-2026
Examination Period :	Mock
Module Code :	MA3508/MA4508/MAT508
Examination Paper Title :	Time series
Duration :	2 hours

Structure of Examination Paper:

- There are **FOUR** pages.
- There are **THREE** questions in total.
- There are no appendices.
- The maximum mark for the examination paper is 75 and the mark obtainable for a question or part of a question is shown in brackets alongside the question.

Instructions to Students:

- Complete the front cover of any answer books used.
- This examination paper must be submitted to an Invigilator at the end of the examination.
- Answer **THREE** questions.
- Each question should be answered on a separate page.
- Show all your working clearly.
- The use of calculators is permitted in this examination.
- The use of a translation dictionary between English or Welsh and another language, provided that it bears an appropriate departmental stamp, is permitted in this examination.

Question 1

(i) State the Mercer expansion and the Karhunen-Loeve expansion.

[2]

(ii) Show that the correlation function $\rho(t)$ of a stationary process is positive semidefinite.

[3]

(iii) Write the R code for computing the 4-step ahead forecast with its 95% confidence interval for the time series 3.2,1.3,4.5,2.7 from a stationary in wide sense process with the covariance function $R(k) = 1.6e^{-0.7|k|}$.

[3]

(iv) Let a process x_t satisfy $x_t - 1.3x_{t-1} = \epsilon_t + 0.8\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Identify if x_t is invertible or not. Give a reason for your answer.

[2]

(v) Consider the causal AR(3) process $x_t = x_{t-1} - 0.5x_{t-2} - 0.2x_{t-3} + \epsilon_t$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$ of this AR(3) process.

[6]

(vi) Suppose that the average precipitation in Paris in successive months was approximated by the AR(2) process $x_t - 11 = 0.5(x_{t-1} - 11) - 0.4(x_{t-2} - 11) + \epsilon_t$, where ϵ_t is Gaussian white noise with zero mean and variance 0.6. Suppose that the last successive observations are 13, 10, 15, 14. Find the predictions for the next two months, the mean square error of predictors, and the confidence intervals with confidence level $1 - \alpha$.

[5]

(vii) Find at least two values of parameter c such that the stochastic process x_t satisfying

$$x_t = -3cx_{t-1} + 2cx_{t-2} + 6c^2 x_{t-3} + \varepsilon_t, \quad t = 0, \pm 1, \ldots,$$

is non-stationary, where ϵ_t is a white noise process with $\mathrm{E}[\epsilon_t] = 0$ and $\mathrm{Var}\,[\epsilon_t] = \sigma_\epsilon^2$.

[4]

Question 2

(i) Give the definitions of an ARMA(p, q) process with zero mean and an ARMA(p, q) process with mean μ.

[2]

(ii) State and prove the condition of causality of the AR(2) process in terms of coefficients.

[4]

(iii) Write the R code for simulating a realization of Gaussian process with the mean function $m(t) = 1.3/(1+t^2)$ and the covariance function $R(t, s) = e^{-|t-s|} + e^{-2|t-s|}$ at points $0.3, 0.7, 0.9, 1.6$.

[3]

(iv) Is the covariance function $R(k) = \frac{5}{(1+|k|^3)^{2/7}}$, $k \in \mathbb{Z}$, short-range or long-range dependent? Explain your answer.

[2]

(v) Consider the time series $x_1 = 62$, $x_2 = 67$, $x_3 = 63$ and assume that the observations correspond to a stationary normal process with covariance function $R(k) = 2/\sqrt{1+k^2}$. Compute the best linear predictor for the next observation $\hat{x}_4$ and also the prediction bounds for x_4 with confidence level $1 - \alpha$.

[6]

(vi) A shop sells books. Sales of books (in units of 100) over the past 12 weeks are $x_1, \ldots, x_{12}$ with $x_{12} = 6.2$. The shop believes that the AR(1) process $x_t - 5 = 0.7(x_{t-1} - 5) + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is Gaussian white noise with zero mean and variance 1.44, is relevant for forecasting. Forecast sales for weeks 13, 14, 15, and 16 and give an expression for the confidence interval of future sales for each of the weeks with confidence level $1 - \alpha$.

[4]

(vii) Find the covariance function for a discrete weakly stationary process x_t with spectral density $f(\lambda) = 1 + |\lambda|$, $\lambda \in [-\pi, \pi]$.

[4]

Question 3

(i) Give the definition of the n-order stationary process and the definition of the Wiener process.

[2]

(ii) Let x_t be a continuous stationary process with zero mean and the covariance function $R(t)$. Suppose that this process was observed at points $t_1, t_2, \ldots, t_n$, where $t_i < t_{i+1}$. Derive the linear predictor of this process at point t with smallest MSE, $t > t_n$.

[5]

(iii) Write the R code for simulating a realization of length 160 from the ARMA(3,1) process defined by the difference equation $x_t - 0.7x_{t-1} + 0.4x_{t-2} - 0.1x_{t-3} = \epsilon_t + 0.5\epsilon_{t-1}$ with $x_1 = 2.1$, $x_2 = -1.2$ and $x_3 = 1.7$, where ϵ_t is white noise with zero mean and variance $\sigma^2 = 1.7$.

[3]

(iv) Let a process x_t satisfy $x_t + \frac{8}{3}x_{t-1} - x_{t-2} = \epsilon_t - \frac{5}{8}\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Identify if x_t is causal or not. Give a reason for your answer.

[3]

(v) Consider the time series $x_k = 1.4\sin(6\pi k/23) + 0.6\sin(4\pi k/13) + 2\sin(3\pi k/11)$, $k \in \mathbb{Z}$, and the filter with spectrum $f(\lambda) = 1_{[0.25\pi, 0.3\pi]}(\lambda)$, $\lambda \in (0, \pi)$. Find the filtered series and its spectrum.

[3]

(vi) Suppose that $x_1, \ldots, x_n$ is a time series from an invertible MA(1) process $x_t = \epsilon_t + b_1\epsilon_{t-1}$, where ϵ_t is white noise. Suppose also that the estimator of the mean of the time series is $\bar{X}_n = 2.7$ and the estimator of the covariance function is $\hat{R}(0) = 4.8$ and $\hat{R}(1) = 1.2$. Estimate parameters of this process: the coefficient b_1 and the noise variance $\sigma^2 = \mathrm{Var}(\epsilon_t)$.

[4]

(vii) Find the variance and correlation function of the AR(2) process $x_t = 0.8x_{t-1} - 0.4x_{t-2} + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is white noise with zero mean and variance $\sigma^2 = 1.3$, using Yule-Walker equations.

[5]